

Research Statement

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1. Research Vision

My research focuses on **AI-assisted software maintenance**, with the goal of delivering **issue-free software on the versions people actually use**. I pursue work in legacy system monitoring, technical debt management, continuous maintenance, cybersecurity maintenance, resource optimization, and software sustainability, in close collaboration with industry. I also train the next generation of computing professionals through teaching and research mentoring.

At its core, the program designs intelligent, data-driven methods and tools that help developers anticipate risk, propagate safe changes, reuse large-scale datasets, and evolve software efficiently—integrating software analytics, machine learning, and systems engineering for trustworthy and scalable software ecosystems.

2. Major Contributions

A. Automation for software evolution at scale. In doctoral work at the University of Saskatchewan (advisors Prof. Chanchal K. Roy and Prof. Kevin Schneider), I developed Baczer, ReBack, BackSlice, and BackTrans to analyze, recommend, and adapt backporting changes. These systems combine repository mining, machine learning (including CNNs and LLMs), and program analysis, and have been evaluated on large open-source ecosystems. Results appear in venues including ICSE, ICPC, MSR, CASCON, ASE Journal, and related software engineering outlets.

B. Scientific workflow data reusability. At the Plant Phenotyping and Imaging Research Centre / Global Institute for Food Security, I developed RISP and RISPs for intermediate data reusability in scientific workflow management on Hadoop/Spark platforms, supporting scalable, reusable research data pipelines.

C. Industry-connected AI automation. As AI Automation Consultant with Cloud Linux Software, Inc., and as Living Skies Postdoctoral Fellow (NSERC CREATE SOAR / related Alliance work), I apply automated backporting and AI-assisted maintenance research to production software-maintenance challenges.

3. Prospective Research Themes

Theme 1: Intelligent release engineering and quality assurance. Machine-learning and hybrid analytical methods to forecast backport risk, estimate quality impact, and recommend testing or remediation pathways for stable releases.

Theme 2: AI-assisted developer workflows and technical debt. Tools and studies connecting issue reports, pull requests, commits, and review discussions to risk-aware release planning and maintainability outcomes.

Theme 3: Industry-connected applied software engineering. Partnerships on legacy modernization, secure maintenance, and automation that generate both scholarly and translational impact.

4. Training, Funding, and Impact

A central goal is to train undergraduate and graduate students in empirical software engineering, reproducible research, and applied software analytics. I pursue staged funding through opportunities such as NSERC Discovery, NSERC Alliance, Mitacs, and institutional seed support, building on existing postdoctoral and industry collaborations.

Over time, I aim to sustain a high-quality publication and artifact pipeline, strengthen academic–industry collaborations, and contribute practical value to software-intensive organizations while integrating research-informed modules into teaching.